

PREMIER UNIVERSITY, CHITTAGONG
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CSE)

PROJECT REPORT

COURSE NAME:
BIG DATA ANALYTICS (CSE 4345)

PROJECT NAME:
EpiScope: A Reproducible District Surveillance and Response-Planning Prototype

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Abstract:

EpiScope demonstrates an inspectable path from public reference sources and seeded synthetic health feeds to district-level analytics and a coordinated D3 dashboard. The prototype responds to a teaching brief for district surveillance, short-term forecasting and response planning, where heterogeneous sources arrive at different times, patient identifiers must be protected, a finite budget must be shared and nontechnical users must understand uncertainty. Apache Spark validates and de-identifies encounter records, applies sixteen quality rules with quarantine reason codes, retains bronze provenance through checksums and batch identifiers, and constructs Parquet marts partitioned by epidemiological year. Checksum-based incremental ingestion makes repeated files a no-op, and late or resubmitted reports update their actual onset weeks. The generator produced 25,000,000 unique encounters and 25,500,523 physical records at scale factor 1.0 across 64 districts and 640 synthetic facilities; 874,178 rows were quarantined and 24,626,345 encounters were accepted. The pipeline was timed at three scale factors, and measured experiments show that a broadcast join and an explicit partition predicate reduced query time in the recorded environment. Outbreak detection uses a weekly CUSUM variant, compared with a seasonal z-score, evaluated on report-availability vintages; at threshold 4 it reached 97.9% measured sensitivity with 9.49 false signal-weeks per district-year. Forecasting uses direct ridge models for horizons of one to four weeks, evaluated over 52 rolling origins against a seasonal-naive baseline, with MAE and nominal-80% interval coverage reported even where the results are unfavourable. Forecasts are converted into resource demand using explicit teaching assumptions, and a linear programme allocates resources under budget, equity, central-stock and transport constraints; five budgets and three equity settings are compared against a constrained population baseline. A seasonal calendar suggests lead times for pre-positioning supplies and starting campaigns. Privacy is addressed through salted hashing, age bands, suppression of counts one to four together with their scores, and aggregate-only export to the browser. The dashboard links a map, forecast timeline, seasonal heatmap, allocation chart and warning table. This report distinguishes measured evidence from design assumptions and incomplete requirements, including the use of public population projections, in-sample prediction intervals, non-ACID year staging and the absence of a formal usability study. Synthetic results do not establish operational surveillance accuracy.

Keywords: Apache Spark; synthetic surveillance; CUSUM; rolling-origin forecasting; resource allocation; D3.js.

Repository URL : *https://github.com/SB-Abhi1/BDA-EpiScope*
Presentation URL : *https://drive.google.com/file/d/1imK_dGQrPre40pLm3VUMRmqRg-ZkajXz/view?usp=drive_link*

I. INTRODUCTION AND PROBLEM ANALYSIS

The teaching brief asks for a district surveillance, short-term forecasting and response-planning platform. The engineering problem is not simply predicting disease counts: heterogeneous sources arrive at different times, identifiers must be protected, a finite budget must be shared and nontechnical users must understand uncertainty. The prototype therefore treats provenance, evaluation and disclosure control as part of the system rather than optional reporting steps.

Decision users include Ministry policymakers, surveillance epidemiologists, district officers, hospitals, vector-control teams, patients and regulators. Their competing needs motivate five decisions: an adjustable detection threshold (C1), district aggregation and suppression (C2), explicit equity constraints (C3), a reproducible daily-batch path (C4), and an interpretable small forecast model with a strong baseline (C5). No stakeholder interviews are claimed.

II. CEP ATTRIBUTES AND DESIGN TRADE-OFFS

P1. Distributed Spark processing, columnar storage, temporal evaluation, optimisation and visual design combine several technical domains.

P2. Sensitivity versus false alerts, detail versus privacy and efficiency versus equity create conflicting objectives.

P3. Thresholds, lags, planning conversion and fairness floors are tested or disclosed rather than treated as uniquely correct.

P4. Delayed reports, resubmission, orphan references and administrative backlogs are explicitly injected.

P5. The combined pipeline needs engineering choices; this prototype does not claim compliance with a complete national health-data standard.

P6. National and district users, hospitals and the public have different needs; the policy brief communicates assumptions without model jargon.

P7. Ingestion affects cases, forecasts affect demand and optimisation affects the dashboard. A failure can propagate across the full chain.

C1 is examined through a two-method threshold curve. C2 suppresses small counts and their derived scores before export and omits exact parent totals. C3 compares five budgets and three equity settings against a constrained population baseline. C4 uses checksum-based incremental batch ingestion rather than a more expensive streaming path; no sub-five-minute freshness claim is made. C5 exposes interpretable lagged features and reports actual error versus a seasonal baseline; no unsupported accuracy win is asserted.

III. ARCHITECTURE AND DATA ENGINEERING

The flow is sources -> bronze -> silver -> gold -> analytics/serving -> D3. Bronze stores unchanged encounter files plus checksums, source paths, batch IDs and load time. Reference/facility/stock inputs are snapshotted by checksum. Silver uses a typed schema, quality rules, salted patient hashing and age bands. Parquet partitions use the ISO epidemiological year of onset week, balancing limited file counts with year-local revisions. Gold contains weekly counts, reporting vintages, latest stock, quality and model outputs. A local persistent Hive catalogue registers the main Parquet tables.

The Spark registry join is broadcast. Record-key windows select the most recent report and ingestion timestamp for a resubmitted encounter. New file checksums are processed incrementally; the same checksum becomes a no-op. Existing records are searched for the resubmitted IDs so an onset-year correction touches both relevant year partitions. Complete year outputs are staged before replacement. Gold is recomputed from the resulting silver snapshot so late reports update their actual onset weeks. This simple design trades recomputation for correctness and remains non-transactional across a multi-partition failure.

The public boundary source is geoBoundaries/BBS/OCHA. Weather is daily Open-Meteo reanalysis for 2021–2025 at 64 district representative coordinates, with raw responses cached. Population totals come from a secondary GIS layer labelled BBS 2022. District age/sex shares are USCB 2022 projections based on 2011 census data, calibrated to those totals. This is a documented public-source equivalent allowed by the brief, not observed BBS2022 age-sex counts. Boundary name aliases are explicitly crosswalked. Coordinates are vertex-mean representatives, not exact area-weighted centroids.

IV. GENERATOR, QUALITY, LINEAGE AND PRIVACY

The seed is 91. Disease-specific seasons, district population/geography and lagged climate drive the expected counts. Gamma-Poisson sampling provides overdispersion. The generator normalises to the requested volume, expands encounter rows in Spark and adds approximately 2% exact resubmissions. It creates 640 facilities and daily logistics snapshots. Thirty-two local events and a nationwide ILI event expanded into 64 district components yield 96 evaluation event records; the latter are not 64 independent national epidemics. Event labels are isolated from the detector and model kernels.

A facility silence/backfill episode preserves onset dates while delaying the corresponding reports. About 0.5% of rows have unknown facilities, and about 1% have invalid age or sex values. Weekends reduce report-day arrivals by shifting reports, capped at the maximum 21-day delay. A pinned Bangladesh holiday calendar shifts report arrivals, subject to the delay cap. Admission probabilities are higher during injected outbreaks and differ by disease. A true three-week reporting silence is enforced for the injected facility, followed by a backlog release.

Sixteen rules cover valid JSON, required encounter/patient keys, facility/district membership, age/sex, ICD-10, parseable and ordered dates, report delay, booleans, outcome and within-batch duplicates. On the full base load, input rows were 25,500,523, quarantined rows 874,178, and accepted encounters 24,626,345. Rules can overlap; failure counts must not be added as disjoint defects.

Quarantine retains reason codes and de-identified fields. The local salt is outside Git, generated with restrictive permissions, and reused on future loads. Patient IDs are salted SHA-256 hashes before silver and exact ages become bands. Weekly gold rows contain exact contributing batch lists, model series carry input batch lists, and a dataset manifest traces derived summaries to loaded batches. Gold snapshots have monotonic snapshot IDs; readers select the latest logical version, preventing retained older Parquet files from

duplicating derived rows after a refresh. This is provenance, not an ACID time-travel guarantee.

Before export, observed counts 1–4 become null together with their CUSUM score. No exact national or annual parent totals are supplied that would restore them by subtraction. Client totals are labelled partial or withheld when necessary. Aggregates and derived forecasts can still carry inference risk; production release needs a broader risk assessment. No patient line lists are served, and the HTTP root is restricted to dashboard/.

V. ANALYTICS METHODS

Detection. A one-sided standardized CUSUM accumulates max(0, previous score + z - 0.5), where z uses only earlier seasonal-reference observations. After a signal the accumulator resets. A seasonal z-score is the comparison method. Thresholds 2, 3, 4, 5, 6 and 8 are evaluated. The reference uses weeks 48–56 in the past when available, otherwise a short guarded baseline. This is a documented weekly CUSUM variant, not a claim of an exact EARS C2 implementation [1].

An injected event matches a signal from six days before its start (weekly-bin allowance) until fourteen days after its end. Sensitivity is matched event records / 96; delay is nonnegative days from start to the first match, with weekly resolution. False alerts count signal-weeks outside all event windows, divided by observed years × 64 districts across all diseases. Successive signal weeks are counted separately, not collapsed into alert episodes. For each issue week the detector reconstructs report availability, estimates current-week completeness from mature prior weeks and scores only known reports. Facility-day backfills after at least seven days of silence, at least ten reports and at least half aged seven days are held out of current automatic alert counts. They remain in revised totals and forecast training. Prefix-invariance tests ensure future reports cannot alter earlier scores. Real-world validation is not claimed.

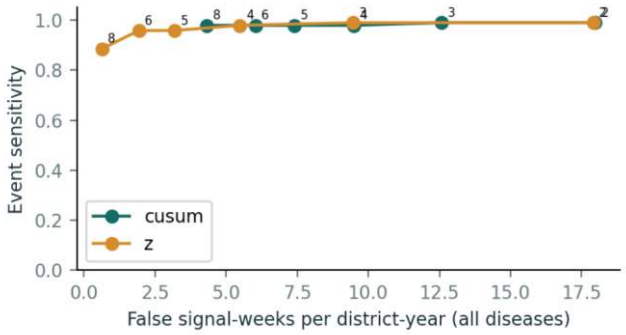


Fig. 1. Measured reporting-vintage sensitivity versus false signal-weeks; labels are thresholds.

Forecasting. Spark groups by district/disease and fits direct log-count ridge models separately for horizons 1–4. Features are recent counts, a four-week mean, the comparable previous-year target week, annual sine/cosine terms and rainfall/temperature lagged six weeks from origin. Nine features and fixed ridge regularisation keep the method inspectable. Training labels do not extend beyond the origin. No future observed weather is used.

Fifty-two rolling origins are evaluated per available district/disease. An aggregated onset/report-week cube reconstructs training counts from reports available at each issue date. The final incomplete onset week is excluded. A seasonal-naive baseline uses the corresponding prior-year week. We report MAE, nominal-80% interval coverage and the 80% interval score. The interval half-width is the 80th percentile of absolute in-sample training residuals; it is not conformal calibration and can under-cover during shifts. Forecast score uncertainty and baseline results are retained even when unfavourable.

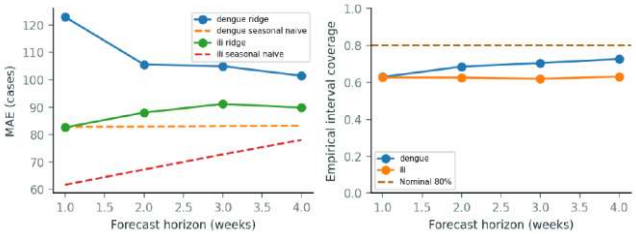


Fig. 2. Forecast error against the seasonal baseline, and empirical coverage of nominal 80% intervals.

Demand conversion. Sum the upper forecasts over four weeks. Per case, assumed bed-days are 0.90 for dengue, 0.21 for diarrhoea and 0.28 each for ILI/malaria. Saline, tests, ORS and antimalarial conversions are explicit in analytics/run.py. Available beds become bed-days over 28 days after subtracting latest occupied beds. These are teaching assumptions, not clinical estimates. Summing district upper bounds is a conservative scenario; it is not a calibrated national quantile.

Allocation. Let g=max(forecast demand-current stock,0). Variables x and u are allocation and unmet demand. Minimise sum(w_d u_dr), subject to u>=g-x, sum(c_r x_dr)<=budget, 0<=x<=g, a capped floor x_dr>=min(g_dr, alpha p_d sum(g_r)), national central-stock caps and division shipment capacity. The central reserve is assumed to be 65% of unmet units per resource; division capacity is 70% of total local unmet units or the floor if larger. Floors, costs and capacities are explicit synthetic planning inputs. Weights are 1 + district population / mean district population, not independently validated severity scores. HiGHS solves the LP [4].

A floor whose cost exceeds budget returns an infeasible result. Fractional quantities are planning estimates; procurement needs integer refinement. The population baseline obeys the same budget, national-stock, transport and unmet-need caps, though it need not spend the full budget. Five budgets (15–90% of gap cost) and three equity settings (0, 10, 20%) expose policy sensitivity. Mixed resource units in the objective also require operational preference calibration.

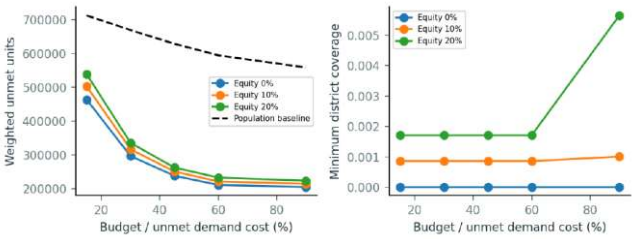


Fig. 3. Budget/equity scenarios; weighted objective and minimum district coverage are both shown.

Calendar. Remove a linear trend from division/disease weekly series, average detrended counts by ISO week, and identify the peak. Pre-position four weeks before it and begin campaigns eight weeks before it. These are explicit lead-time

assumptions. The calendar represents estimated recurring seasonality, not a prediction of the next novel epidemic.

VI. EVALUATION AND PERFORMANCE

Disease/h	MAE	Naive	80% cover
dengue / 1	122.9	82.9	63.0%
dengue / 4	101.5	83.2	72.7%
ili / 1	82.7	61.7	62.7%
ili / 4	89.9	78.1	63.2%

Table I reports selected horizons; the full per-disease/horizon metrics include all four diseases. Accuracy is an empirical result on generated patterns, not epidemiological external validity. The generator and model share seasonal/climate structure, so synthetic success can overstate performance on real reporting systems.

At CUSUM threshold 4, measured reporting-vintage sensitivity was 97.9%, false signal-weeks per district-year 9.49, and median matched-event delay 0.0 days. This default is an inspectable example operating point, not a stakeholder-approved policy.

SF	Unique	Gen s	Lake s	Model s	JSON s	Total s
0.25	6.25M	65.6	184.4	78.3	1.1	329.3
0.5	12.50M	124.9	231.9	82.4	1.3	440.5
1.0	25.00M	273.5	525.6	106.0	1.0	906.1

The measured SF1 generator/lake runtime was 799.1 s. SF1 analytics took 106.0 s separately; export timing is recorded in the evidence bundle. The environment had 9.7 GiB host RAM, local[2] Spark execution and a 3 GiB configured driver heap. Source acquisition is cached/network-dependent and excluded. Analytics and privacy export were also run at each scale. A final clean SF1 build records the completed generator, lake, analytics and export in one run; its timing and the earlier staging-file failure are retained in the evidence.

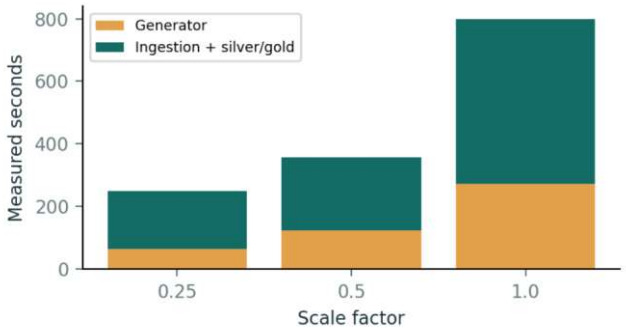


Fig. 4. Actual generator and lake times at the three required scale factors; plot shows generator/lake only; Table II includes analytics and export at every scale.

Experiment	Variant	Time s
Registry join	merge	9.38
Registry join	broadcast	2.17
Two aggregates	uncached	2.38
Two aggregates	cached including materialisation	12.97
Partition pruning	no partition predicate	1.12
Partition pruning	partition predicate	0.40

Join timings use three repetitions per hint; medians are reported. Cached aggregation includes its materialisation cost. Experiments are sequential and affected by warm caches and order; they are not controlled hardware-isolated benchmarks. Identical query checksums verify comparable results. The partition experiment compares an ISO-year expression on onset dates with the equivalent explicit partition predicate; the latter prunes unrelated years. Negative or negligible gains are reported rather than discarded. Physical plans and real Spark UI captures are included.

The late-data integration fixture passed: True. It verified one accepted key, duplicate rejection, bad-age/date quarantine, repeated-checksum no-op and a corrected onset week without duplicate inflation. Core tests also verify budget/floor constraints, infeasibility handling, forecast prefix invariance, CUSUM prefix invariance and small-count suppression. Tests establish selected invariants; they do not prove all hidden-drop behaviours.

A separate 14-day schema-compatible drop added 3,584 valid encounters, 20 resubmissions and 10 invalid-age records. It used an unexpected filename, late onset dates and a new-year partition. Repeat ingestion was a no-op and forecasts extended to 2026-02-02. This is a simulation, not the instructor's actual hidden file.

VII. DASHBOARD DESIGN AND USABILITY

D3 v7.9.0 supplies the map, forecast timeline with brush, seasonal heatmap, allocation chart and warning table. A sequential teal scale communicates incidence, gold marks forecasts/signals and explicit gaps/hatching distinguish suppression from zero. District and disease controls coordinate the historical views and demand-proportional allocation attribution. A historical brush filters the current forward allocation out when its issue origin is outside the selected window; the interface explains that historical allocation scenarios are unavailable. All views respond to the shared selections.

Only aggregated JSON and simplified district geometry are loaded. Measured combined payload is 5,531,910 bytes, below 10 MiB; 4,758 observed cells are withheld. Browser checks cover SVG rendering, filter changes, table coordination, what-if updates and page errors. Real users have not yet completed a usability study; no invented findings are supplied. Mobile layout is responsive, but there is no offline service worker or formal accessibility audit.

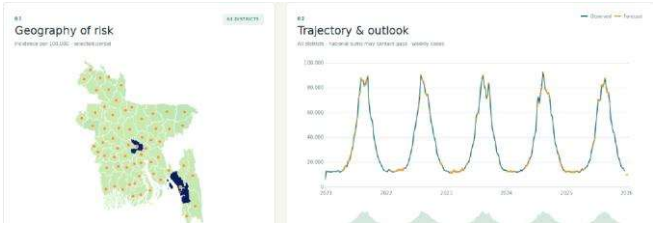


Fig. 5. Actual browser rendering of the district map and forecast timeline. Full screenshots are in the evidence package.

VIII. ETHICS, SOCIAL AND SUSTAINABILITY CONSIDERATIONS

Misleading outbreak signals can stigmatise districts or redirect scarce supplies. A model trained on synthetic patterns cannot justify claims about real people. The interface labels synthetic inputs, displays uncertainty and uses review language rather than outbreak confirmation. Source/license review, purpose limitation, access control, retention rules and local health expertise are prerequisites for any real deployment.

Fairness is not guaranteed by a single floor. We show both weighted unmet units and minimum district coverage, disclose population-driven weights and compare alternative budgets. Patient privacy is strengthened by minimising release and suppressing linked outputs, but hashing is pseudonymisation rather than anonymity. A public dashboard would need a fuller inference-risk assessment. Efficient local processing and cached weather reduce repeated compute/network use; no measured carbon saving is claimed.

IX. LIMITATIONS AND FUTURE WORK

- Population age-sex denominators use disclosed USCB projections as a public-data equivalent under the brief's alternatives allowance. Estimates are not observed BBS2022 age-sex counts. Verify intended deployment sources and terms before operational use.
- The generator includes a pinned Bangladesh holiday calendar and outbreak-dependent admission changes. Its output remains a teaching simulation rather than an epidemiologically validated model.
- Detection uses weekly report vintages, an estimated completeness factor and a backfill guard. Broad event-matching windows and correlated nationwide event components affect metric interpretation; independent real-world validation is still needed.
- Intervals use in-sample residuals and need calibration. Demand factors, population weights, central reserve and transport capacities are assumptions; allocation uses fractional units.
- Year staging is not an ACID multi-table commit. Concurrent writers, crash recovery, durable object storage, streaming, time travel and role-based access are not implemented.
- The compute path is evaluated at three scales in the recorded environment. Formal two-user usability testing and independent clean-machine/Docker validation remain outstanding. Native Windows full-scale execution has not been verified.

The prototype supports reproducible investigation of surveillance and planning methods. Operational deployment requires verified sources, prospective evaluation and appropriate governance.

REFERENCES

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[3] Open-Meteo, "Historical Weather API." https://open-meteo.com/en/docs/historical-weather-api

[4] SciPy, "scipy.optimize.linprog / HiGHS." https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linprog.html

[5] geoBoundaries, "API." https://www.geoboundaries.org/api.html ; BGD ADM2 metadata cached in data/reference/.

[6] U.S. Census Bureau, "Subnational Population by Sex, Age, and Geographic Area." https://www.census.gov/data/tables/time-series/demo/international-programs/subnationalpopulation.html

[7] D3, version 7.9.0, ISC license. https://github.com/d3/d3

[8] ArcGIS Bangladesh, district demographic layer labelled BBS2022, ADB005/CRIIPS/MapServer/355. Source endpoint and checksums in provenance files.

[9] User-provided "CEP EpiScope" teaching brief, CSE 4345 Big Data Analytics, Spring 2026, 12 pp. Instructor and institution as stated in supplied brief. No government endorsement implied.

Prepare earlier.
Allocate transparently.

To: Director, Disease Control & Preparedness Unit

Purpose: assess a district-level surveillance and response-planning prototype.

Decision in one sentence

Use EpiScope as an inspectable training and planning prototype; require verified demographic inputs, prospective alert validation and calibrated forecast uncertainty before any operational pilot.

What the prototype provides

The platform joins district boundaries, public weather and population estimates with synthetic health and stock records. It detects unusual patterns, forecasts weekly cases one to four weeks ahead, translates forecasts into resource demand and compares allocation choices. An interactive dashboard links a map, case timeline, seasonal heatmap, allocation chart and ranked warning table.

Evidence	Measured result
Generated encounter volume	25,000,000 unique + deliberate resubmissions
Districts and facilities	64 districts; 640 synthetic facilities
Lake processing	525.6 s in the preparation environment
Privacy boundary	No patient records in browser; counts 1-4 withheld

Three recommended actions

1. Verify before acting. Replace or formally approve the alternative demographic projections. Review alert timing using data as it was actually reported, including silence and backlog episodes. Expected benefit: fewer decisions based on source mismatches or administrative artefacts.

2. Choose equity explicitly. Compare budget and equity scenarios and record the chosen floor, stock limits and transport assumptions. Expected benefit: a transparent explanation of who receives resources and which demand remains unmet.

3. Keep uncertainty visible. Show district prediction ranges, measured interval coverage and the date of the latest data. Obtain local review before dispatch. Expected benefit: decisions that account for forecast error rather than treating a point estimate as a promise.

How to read the evidence

Map: darker colour means higher disclosed incidence per 100,000. Gold markers are signals for review. Hatching indicates withheld data, not absence of disease.

Timeline: teal shows revised onset-week observations; gold shows forecasts. District shading is a nominal 80% prediction interval. Missing segments can mean small-count suppression.

Allocation: compare forecast-derived demand, latest stock and proposed allocation. The disease filter attributes a joint plan in proportion to forecast demand. A higher budget may stop helping once central stock or transport limits bind.

Quality: inspect invalid and duplicate records, reporting delays, model performance against the seasonal baseline and export freshness.

A policy scenario to discuss, not an endorsed budget

At a budget equal to 45% of estimated unmet-demand cost and a 10% equity setting, weighted unmet units are 251,431, compared with 628,466 under the population baseline. This comparison uses synthetic demand, assumed conversion factors and assumed national/transport capacity; it does not establish real-world savings.

Confidence and limitations

All encounter and stock records are synthetic. The population age/sex profiles are public projections calibrated to a secondary BBS-labelled district-total layer. Detection evaluation uses report-availability vintages and a generic backfill guard. Forecast interval coverage is measured, not guaranteed to equal 80%. Allocations are fractional planning quantities, not approved shipments. The implementation is a local batch prototype; real operational and human-usability validation remain outside its synthetic evaluation.

Data protection assurance

The browser receives district-level aggregates only. Direct patient identifiers do not pass into silver, the hashing salt remains outside source control, and counts one through four are suppressed together with their signal score. Exact parent totals are omitted to reduce subtraction attacks. Public release would still require governance, access controls, source-license review and a broader disclosure-risk assessment.

Accountability

An alert requests investigation; it does not confirm an outbreak. A suggested allocation supports a documented human decision. Keep the input snapshot, selected policy settings and final decision together so that the outcome can be reviewed.

APPENDIX B / CONTRIBUTION MATRIX

The role allocation assigns one third to each member, totalling 100%. L = lead; S = support. Signatures confirm the contribution record; Git statistics are drawn from the actual repository history.

Work area	Dipton Dey	Shaishob Boidya	Manik Dey
Problem analysis / proposal	L	S	S
Generator / public sources	L	S	S
Ingestion / lake / incremental	L	S	S
Quality / lineage / privacy	L	S	S
Detection / evaluation	S	L	S
Forecasting / demand	S	L	S
LP / calendar	S	L	S
Serving / D3 dashboard	S	S	L
Scaling / Spark optimisation	L	S	S
Report / policy brief	S	S	L
Video production	S	S	L
Contribution share	33 1/3%	33 1/3%	33 1/3%
Git commits / lines	_____	_____	_____
Signature	_____	_____	_____

Contribution evidence

Contribution percentages describe the team's allocation. Individual commit and line statistics are separate measures and must correspond to the repository history.

APPENDIX C / INDIVIDUAL REFLECTIONS

Dipton Dey:

My focus in the team role plan is the data engineering path: synthetic generation, public reference inputs, bronze and silver processing, quality checks, lineage and incremental ingestion. The main engineering lesson is that a large record count is useful only when the meaning and provenance of those records remain inspectable. Raw snapshots and checksums support reproducibility, while quarantine reason codes make rejected records explainable.

The design uses Parquet partitions by epidemiological year and checksum-based repeat-file detection. This balances simple local execution with selective reprocessing. A transactional table format would provide stronger atomicity, but it would also add deployment complexity. The documented restart limitation therefore matters when deciding whether the prototype is suitable beyond a teaching environment.

The relevant verification evidence includes the SF1 record totals, the repeat-file no-op check, onset-week correction and the measured broadcast-join and partition-pruning comparisons. These measurements describe the recorded environment; reproduction is necessary before generalising them to other hardware. My next improvement would be to strengthen recovery around partition replacement and validate it with an interrupted-write scenario.

Shaishob Boidya:

My focus in the team role plan is the analytical layer: outbreak signals, weekly forecasting, rolling-origin evaluation, resource demand, constrained allocation and the seasonal calendar. The central lesson is that plausible model output is not enough. Evaluation must reflect what information would have been available at the time of each prediction, especially when disease reports arrive late or are revised.

The prototype compares CUSUM with a seasonal z-score detector and compares direct ridge forecasts with a seasonal-naive baseline. A more complex model could be considered, but the reported weak forecasting performance makes the baseline comparison essential. Nominal 80% intervals also need empirical coverage checks before their upper bounds are used for resource planning. The linear programme makes budget and equity choices visible, although its resource coefficients and capacities remain synthetic assumptions.

The relevant verification evidence includes the threshold trade-off, report-vintage backtests, forecast errors and interval coverage, and feasible budget/equity scenarios against the population baseline. I would prioritise interval calibration and prospective alert validation before an operational pilot.

Manik Dey:

My focus in the team role plan is the serving and communication layer: privacy-filtered exports, the linked D3 dashboard, report coordination and presentation

planning. The main lesson is that a dashboard is part of the analytical system. A clear visual display can still mislead if its date range, uncertainty, suppressed values or underlying assumptions are unclear.

The prototype exports district aggregates instead of patient records and suppresses counts from one to four together with their signal scores. Exact parent totals are omitted to reduce subtraction risk. Five linked views let a reader move between geography, historical cases, seasonality, allocation and warnings. Precomputed allocation scenarios keep browser interaction manageable, while limiting the dashboard to the policies actually evaluated.

The relevant verification checks include district and disease coordination, date brushing, suppression behaviour, JSON size and whether allocation is hidden outside its forecast issue period. Screenshots alone do not demonstrate usability, so a real user walkthrough remains necessary. My next improvements would be keyboard and screen-reader checks, clearer task-based feedback and a documented usability study.

APPENDIX D / REQUIREMENT TRACEABILITY

Status is intentionally explicit. “Implemented” is not equivalent to independently verified, and partial M requirements must be resolved before claiming full compliance. Full paths and evidence notes are also in requirements.csv

ID	Pri.	Status	Implementation / evidence
ING-1	M	implemented	S1-S6 raw snapshots + metadata; demographic source substitution disclosed
ING-2	M	verified	Spark transformations; NumPy used only for compact generator/control computations
ING-3	M	implemented	Parquet ISO-year partitions; persistent Hive catalogue registered; gold recomputed after updates
ING-4	M	verified with restart limitation	Checksum no-op and onset-week correction tested; year replacement not fully atomic
ING-5	S	not implemented	Local filesystem only; object-store/HDFS deployment absent
ING-6	S	not implemented	Batch ingestion only; no Kafka/streaming path
ING-7	C	not implemented	No ACID/time-travel table format
DQ-1	M	verified	16 rules; quarantine reason codes
DQ-2	M	verified	Per-load report in gold table, JSON and dashboard
DQ-3	M	implemented	Weekly exact batch IDs; model series batch lists; dataset manifest for derived summaries
DQ-4	S	not implemented	No patient deduplication across facilities beyond encounter-key deduplication
PRIV-1	M	verified	Salted hash before silver; no direct patient IDs in exports; local secret excluded
PRIV-2	M	verified	Counts 1-4 and score withheld; no exact parent totals; structural/export tests
PRIV-3	S	not implemented	No authentication or role-based detail access
PRIV-4	C	not implemented	No record-level public extracts; k-anonymity/DP implementation absent
AN-1	M	verified with reporting vintages	CUSUM by district/disease in grouped Spark
AN-2	M	verified	Reporting-vintage sensitivity, false signal-weeks, detection delay and threshold curve
AN-3	S	verified	CUSUM vs seasonal z; facility-day backfill guard, injected-episode check and backlog invariance tests
AN-4	M	verified	4 horizons, 4 diseases, lagged weather, parallel Spark
AN-5	M	verified; intervals not calibrated	52 rolling origins, vintage-filtered training, MAE, seasonal baseline, nominal-80 coverage and interval score
AN-6	M	implemented; assumptions explicit	Upper-bound forecasts to bed-days, saline, tests, ORS and antimalarials
AN-7	M	verified	HiGHS LP: budget, equity, stock, transport; feasible population baseline
AN-8	S	implemented	5 budgets x 3 equity floors; curves + current example policy
AN-9	M	implemented	Detrended seasonal week component by division/disease; explicit lead-time assumptions
AN-10	C	not implemented	No SEIR scenario module
VIS-1	M	verified	Bundled D3 v7.9.0, no prohibited chart library
VIS-2	M	verified	Five views; real district boundaries, incidence, intervals, heatmap, allocation, warning table
VIS-3	M	verified	District/disease coordinate all views; brush filters historical views and hides forward allocation outside its issue period
VIS-4	M	verified	Aggregate JSON + geometry checked under 10 MiB; no line lists
VIS-5	S	implemented	Threshold plus 15 precomputed budget/equity LP scenarios
VIS-6	S	partial	Color-safe sequential palette, suppression/uncertainty and page freshness; no formal accessibility or user study
VIS-7	C	not implemented	No mobile/offline service worker or animated spatial playback
ENG-1	M	native path verified; Docker untested	Pinned native dependencies, one-command rebuild; external data network/source availability still applies
ENG-2	M	verified	Complete compute timings at all three scales; final clean SF1 rebuild recorded; source downloads cached
ENG-3	M	verified with measured results	Broadcast join and partition-pruning gains; cache regression disclosed; actual plans and Spark UI
ENG-4	S	partial	Core invariants and late-data integration tested; no orchestrated retry DAG